

- An NTP server that will synchronise the time between the client and the AD server (assuming the Kerberos server and AD are running on the same system). This is not a Likewise requirement; it is needed for the Kerberos server, because if the time (system clocks) on the client and on the Kerberos server differs by more than 5 minutes, the Kerberos server will refuse tickets, as a security measure.

How it works

Although a detailed coverage of Kerberos authentication processes is out of scope for this article, let's take a quick look at the basic client-request/server-response sequence.

When a user attempts to log in to the Linux system, the login program obtains the username and password, and passes the credentials to PAM (the Pluggable Authentication Modules system). When the Likewise agent is installed, its library modules and configuration files set it up as the PAM module to use for authentication. Thus, PAM passes the credentials to the Likewise authentication daemon. This, in turn, generates a secret key from the username and password, and then uses the secret key to request a TGT (Ticket-Granting Ticket) from the Kerberos server, which runs as part of the KDC (Key Distribution Centre) on the AD server.

The Kerberos authentication server looks up the client in its database, and then generates a session key (SK1) for use between the client and the Ticket-Granting Server (TGS). Kerberos encrypts the SK1 using the client's secret key. The authentication server uses the TGS's secret key (known only to the authentication server and the TGS), to create a ticket-granting ticket (TGT), and sends it to the client.

Once the Likewise daemon receives the generated TGT, it can ask the TGS for (and receive) service tickets for access to other services on the network (without having to individually authenticate to each service with a username/password).

Installing Likewise

If your Linux distribution does not have Likewise packages in its repositories, download the Likewise GUI installer agent (a Bitrock installer) from the URL mentioned earlier. Right-click the downloaded file and change the permissions to 'executable'. Double-click the installer file, and follow the prompts to install the software. Once the installation is done, the installer displays the message shown in Figure 1.

Getting a desktop to join AD

You need to be the *root* on the Linux desktop system, to run the command-line tool *domainjoin-cli*, which accepts the 'join' option, followed by the arguments *domainName* (Fully Qualified Domain Name or FQDN of the domain to join), and *joinAccount* (a username that is authorised to

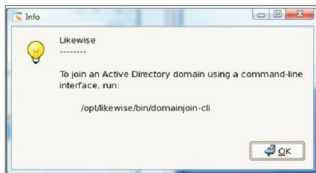


Figure 1: Post-installation message



Figure 2: Joining AD via command-line utility

create computer accounts in the AD domain). The syntax is as follows:

```
/opt/likewise/bin/domainjoin-cli join <domainName> <joinAccount>
```

In Figure 2, which shows the use of the command-line tool: *likewise.example.com* is the FQDN of the AD domain to join.

Administrator is the user account that is privileged to join computers to the domain.

To join a specific Organisational Unit in the AD, use the additional *--ou* switch:

```
domainjoin-cli join --ou <OUName> <domainName> <joinAccount>
```

Example: `domainjoin-cli join --ou ITS likewise.example.com Administrator`